Introduction to Programming(II) Final Term Exam Name: Student ID:

1. Write a simple C++ program to demonstrate how "**short circuit evaluation**" works. You should also explain your program. (10%)
2. Write a simple C++ program to demonstrate "**function overloading**". You should also explain your program. (10%)
3. Write a simple C++ program to explain the "**memory leak**" problem. You should also explain your program. (10%)
4. Explain what **abstract classes** are, and write an abstract class in C++. (10%)
5. Explain what **higher-order functions** are, and write a higher-order function in C++. (10%)
6. Write a simple C++ program to demonstrate the use of **Lambda Expression.** You should also explain your program. (10%)
7. What is the output of the following program? (10%)

```cpp
#include<iostream>
using namespace std;
class Shape {
public:
    double area;
    virtual double calculateArea() { return 0.0; };
    void printArea() { cout << "(Shape) " << area << endl; }
};
class Square: public Shape {
public:
    double side;
    double calculateArea() { return side * side; };
    void printArea() { cout << "(Square) " << area << endl; }
};
void f(Shape & s) {
    s.area = s.calculateArea();
    s.printArea();
}
int main(int argc, const char * argv[]) {
    Square square;
    square.side = 3.0;
    f(square);
    square.printArea();
    Shape shape = square;
    f(shape);
    shape.printArea();
    Shape *sp = &square;
    f(*sp);
    sp->printArea();
}
```

8. What is the output of the following program? (10%)

```cpp
#include<iostream>
using namespace std;
class TwoNumbers {
public:
    int x; int y;
    TwoNumbers(int x, int y) {this->x = x+1; this->y = y;}
    TwoNumbers operator^(const TwoNumbers & tn) const {
        return TwoNumbers(x-tn.x, y*tn.y);
    }
};
int main(int argc, const char * argv[]) {
    TwoNumbers tn1(6, 2); TwoNumbers tn2(5, 3);
    TwoNumbers tn3 = tn1 ^ tn2;
    cout << tn3.x << ", " << tn3.y << endl;
}
```

9. What is the output of the following program? (10%)

```
#include<iostream>
using namespace std;
template<class T, class S>
class Check {
public:
    bool isEqual(T data1, S data2) {
        if (data1 == (T)data2) {
            return true;
        } else {
            return false;
        }
    }
};
int main() {
    cout << boolalpha;
    int i1 = 6, i2 = 8, i3 = 6;
    double d1 = 6.3, d2 = 8.6;
    Check<int, double> c1;
    cout << c1.isEqual(i1, d1) << endl;
    Check<double, int> c2;
    cout << c2.isEqual(d2, i2) << endl;
    Check<int *, int *> c3;
    cout << c3.isEqual(&i1, &i3) << endl;
}
```

10. (1) Figure out the lines that will cause compilation error. (2) After removing the lines that will cause compilation error, what is the output of this program? (10%)

```
01 #include<iostream>
02 using namespace std;
03 class A {
04 private:
05     int a1;
06 protected:
07     int a2;
08 public:
09     int a3;
10     A(int value) { a1 = value+6; a2 = value+7; a3 = value+8; }
11     A(const A &a) { a1 = a.a1+2; a2 = 1; }
12     int fun1() { return (a1); }
13 };
14 class B: public A {
15 private:
16     int b1;
17 protected:
18     int b2;
19 public:
20     B(): A(6) { b1 = 3; b2 = 5;}
21     int fun1() { return A::fun1(); }
22     int fun2() { return (b1+a1); } (X)
23     int fun3() { return (b2+a2); }
24 };
25 int main(int argc, const char * argv[]) {
26     A a = A(7);
27     B b1;
28     B b2 = b1;
29     cout << a.a3 << endl;
30     cout << b1.a3 << endl;
31     cout << b2.a3 << endl;
32     cout << a.fun1() << endl;
33     cout << b1.fun1() << endl;
34     cout << b2.fun1() << endl;
35     cout << a.fun3() << endl;
36     cout << b1.fun3() << endl;
37     cout << b2.fun3() << endl;
38 }
```